The documentation and process conversion measures necessary to comply with this document shall be completed by 25 Month 2019.

INCH-POUND

MIL-PRF-19500/728D
26 September 2019
MIL-PRF-19500/728C
SUPERSEDING
25 October 2010

PERFORMANCE SPECIFICATION SHEET

* SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING, TYPES 2N3724, 2N3725, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for NPN, silicon, switching transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500 and two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type. RHA level designators "M", "D", "P", "L" "R", and "F' are appended to the device prefix to identify devices which have passed RHA requirements.

* 1.2 Physical dimensions. The device package for the encapsulated device type is as follow: (TO-5 and TO-39) in accordance with figure 1 and (UB) in accordance with figure 2. The dimensions and topography for JANHC and JANKC unencapsulated die is as follows: The A version die in accordance with figure 3.

1.3 Maximum ratings unless otherwise specified $T_A$ = +25°C.

| Types | $I_C$ | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|
| | A dc | V dc | V dc | V dc | °C |
| 2N3724, L, UB | 1 | 50 | 30 | 6.0 | -65 to +200 |
| 2N3725, L, UB | 1 | 80 | 50 | 6.0 | -65 to +200 |

| Types | $P_T$ $T_A$ = +25°C (1) | $P_T$ $T_C$ = +25°C (1) | $P_T$ $T_{SP(IS)}$ = +25°C (1) | $R_{\theta JA}$ (2) (3) | $R_{\theta JC}$ (2) (3) | $R_{\theta JSP(IS)}$ (2) (3) |
|---|---|---|---|---|---|---|
| | W | W | W | °C/W | °C/W | °C/W |
| 2N3724 | 0.800 | 5 | N/A | 175 | 35 | N/A |
| 2N3725 | 0.800 | 5 | N/A | 175 | 35 | N/A |
| 2N3724L | 0.800 | 5 | N/A | 175 | 35 | N/A |
| 2N3725L | 0.800 | 5 | N/A | 175 | 35 | N/A |
| 2N3724UB | 0.500 | N/A | 1.5 | 350 | N/A | 90 |
| 2N3725UB | 0.500 | N/A | 1.5 | 350 | N/A | 90 |

(1) For derating, see figure 4, figure 5, figure 6, and figure 7.
(2) See 3.3.
(3) For thermal impedance curves, see figure 8, figure 9 figure 10, and figure 11.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics at $T_A$ = +25°C.

| Types | $h_{FE1}$ $I_C$ = 100 mA dc $V_{CE}$ = 1 V dc | | $h_{FE2}$ $I_C$ = 300 mA dc $V_{CE}$ = 1 V dc | | $h_{FE3}$ $I_C$ = 500 mA dc $V_{CE}$ = 1.0 V dc | | $\lvert h_{fe} \rvert$ f = 100 MHz $V_{CE}$ = 10 V dc $I_C$ = 50 mA dc | | $C_{obo}$ 100 kHz ≤ f ≤ 1 MHz $V_{CB}$ = 10 V dc $I_E$ = 0 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Min | Max | Min | Max | Min | Max | Min | Max | Min pF | Max pF |
| 2N3724, L | 60 | 240 | 40 | | 35 | | 3.0 | | | 12 |
| 2N3724UB | 60 | 240 | 40 | | 35 | | 3.0 | | | 12 |
| 2N3725, L | 60 | 240 | 35 | | 30 | | 3.0 | | | 12 |
| 2N3725UB | 60 | 240 | 35 | | 30 | | 3.0 | | | 12 |

| Types | $V_{CE(sat)2}$ (1) $I_C$ = 100 mA dc $I_B$ = 10 mA dc | | $V_{CE(sat)1}$ (1) $I_C$ = 10 mA dc $I_B$ = 1.0 mA dc | | $V_{BE(sat)1}$ (1) $I_C$ = 10 mA dc $I_B$ = 1.0 mA dc | | $V_{BE(sat)2}$ (1) $I_C$ = 100 mA dc $I_B$ = 10 mA dc | | Switching (saturated) $t_{on}$ See figure 12 | | Switching (saturated) $t_{off}$ See figure 13 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | V dc Min | V dc Max | V dc Min | V dc Max | V dc Min | V dc Max | V dc Min | V dc Max | ns Min | ns Max | ns Min | ns Max |
| 2N3724, L, UB, | | .2 | | .25 | | .76 | | .86 | | 50 | | 60 |
| 2N3725, L, UB | | .26 | | .25 | | .76 | | .86 | | 50 | | 60 |

(1) Pulsed see 4.5.1.

* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level designators.

* 1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", “JANTXV”, and "JANS".

* 1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

* 1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest for JANS quality levels are as follows: "M", "D", "P", "L", “R”, and “F”. For the RHA levels for TXV quality levels are as follows: “R” and “F’.

* 1.5.3 Device type. The designation system for the device types covered by this specification sheet are as follows.

* 1.5.3.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "2N".

* 1.5.3.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "3724".



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* 1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN for this specification sheet.

| | |
|---|---|
| | A blank first suffix symbol indicates a 3 lead metal can package with lead dimension LL = .5 inches (12.70 mm) min and .750 inch (19.05 mm) max. (see figure 1, TO-39). |
| L | Indicates a 3 lead metal can package with lead dimension LL = 1.5 inches (38.10 mm) min and 1.75 inches (44.45 mm) max. (see figure 1, TO-5) |
| UB | Indicates a 4 pad surface mount package, (see figure 2) |

* 1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

* 1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet is "A".

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

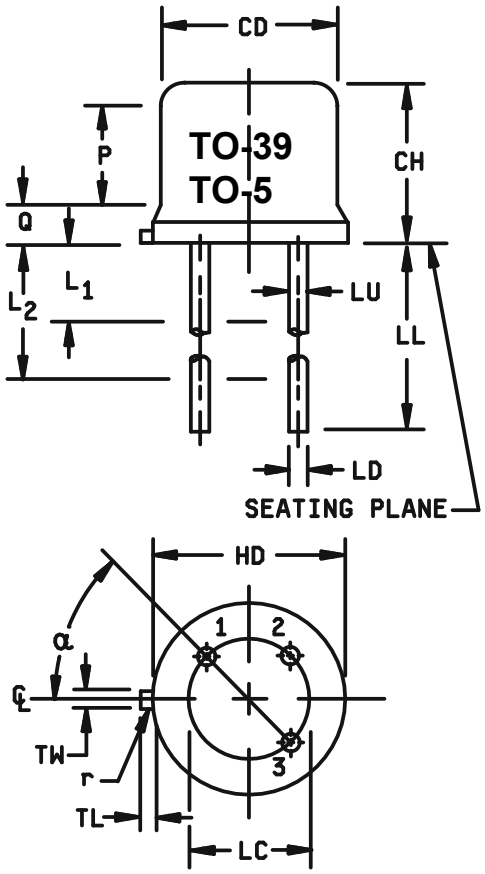
* (Copies of these documents are available online at https://quicksearch.dla.mil.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LC | .200 TP | | 5.08 TP | | 7 |
| LD | .016 | .019 | 0.41 | 0.48 | 8,9 |
| LL | See note 14 | | | | |
| LU | .016 | .019 | 0.41 | 0.48 | 8,9 |
| $L_1$ | | .050 | | 1.27 | 8,9 |
| $L_2$ | .250 | | 6.35 | | 8,9 |
| P | .100 | | 2.54 | | 7 |
| Q | | .030 | | 0.76 | 5 |
| TL | .029 | .045 | 0.74 | 1.14 | 3,4 |
| TW | .028 | .034 | 0.71 | 0.86 | 3 |
| r | | .010 | | 0.25 | 10 |
| $\alpha$ | 45° TP | | 45° TP | | 7 |

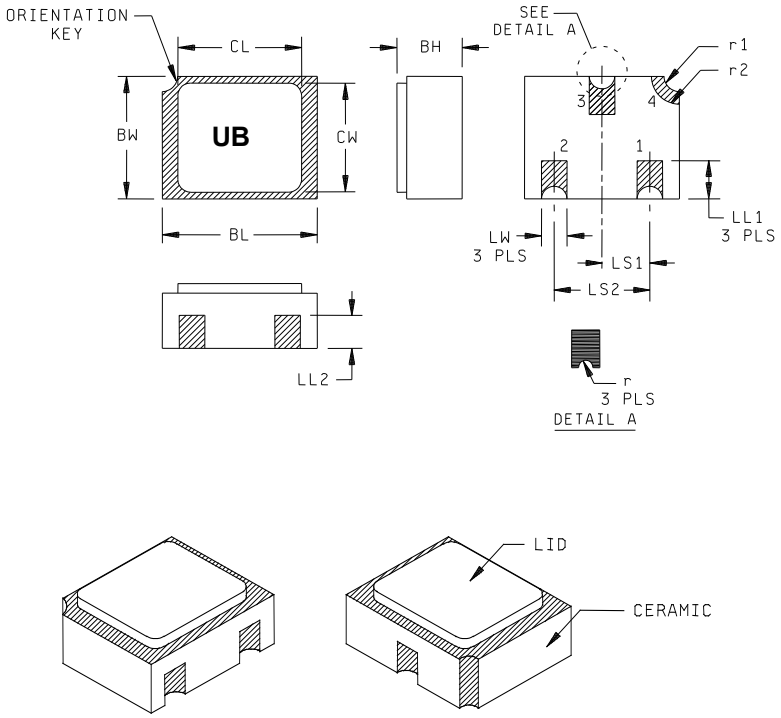



NOTES:
1. Dimensions are in inches.
2. Millimeters equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
8. Dimension LU applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
9. All three leads.
10. The collector shall be internally connected to the case.
11. Dimension r (radius) applies to both inside corners of tab.
12. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.
13. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
14. For L suffix devices (TO-5), dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max. For non-L suffix types (TO-39), dimension LL = .5 inch (12.70 mm) min. and .750 inch (19.05 mm) max.

FIGURE 1. Physical dimensions for 2N3724 and 2N3725 (TO-5 and TO-39).



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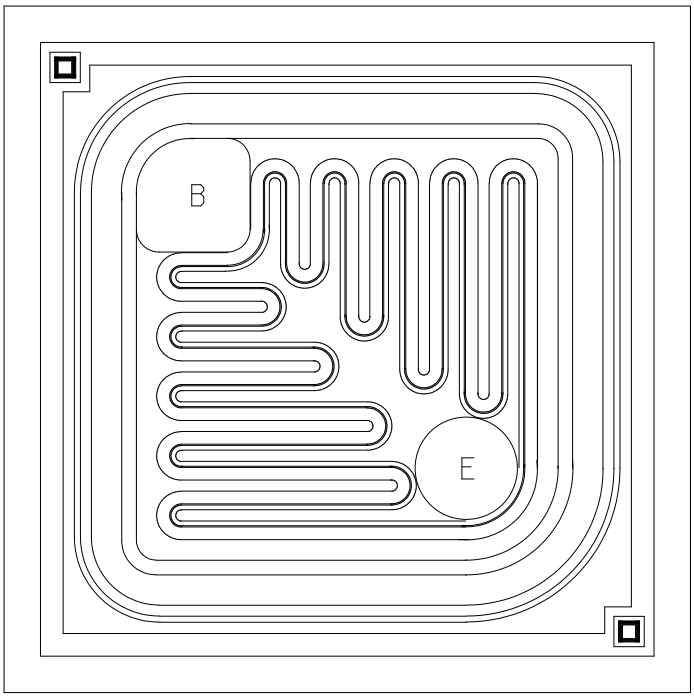

| Symbol | Dimensions | | | | Note | Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | | Inches | | Millimeters | | |
| | Min | Max | Min | Max | | | Min | Max | Min | Max | |
| BH | .046 | .056 | 1.17 | 1.42 | | $LS_1$ | .035 | .039 | 0.89 | 0.99 | |
| BL | .115 | .128 | 2.92 | 3.25 | | $LS_2$ | .071 | .079 | 1.80 | 2.01 | |
| BW | .085 | .108 | 2.16 | 2.74 | | LW | .016 | .024 | 0.41 | 0.61 | |
| CL | .115 | .128 | 2.92 | 3.25 | | r | | .008 | | 0.20 | |
| CW | .085 | .108 | 2.16 | 2.74 | | r1 | | .012 | | 0.31 | |
| LL1 | .022 | .038 | 0.56 | 0.97 | | r2 | | .022 | | 0.56 | |
| LL2 | .017 | .035 | 0.43 | 0.89 | | | | | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters equivalents are given for general information only.
3. Hatched areas on package denote metallized areas
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions, surface mount 2N3724UB and 2N3725UB (UB version).



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NOTES:

| | | |
|---|---|---|
| 1. | Die size: | .025 x .025 inch ±.002 inch (0.635 mm x 0.635 mm). |
| 2. | Die thickness: | .010 ±.0015 inch (0.254 mm ±0.0381 mm). |
| 3. | Base bonding pad: | .004 x .004 inch (0.1016 mm x 0.1016 mm). |
| 4. | Emitter bonding pad: | .004 x .004 inch (0.1016 mm x 0.1016 mm). |
| 5. | Back metal: | Gold, 5,000 Å. |
| 6. | Top metal: | Aluminum, 15,000Å minimum, 18,000 Å nominal. |
| 7. | Back side: | Collector. |
| 8. | Glassivation: | $Si_3N_4$, 8,000 Å nominal. |

FIGURE 3. JANHC and JANKC die dimensions.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

| | |
|---|---|
| PCB | Printed circuit board. |
| $R_{\theta JA}$ | Thermal resistance junction to ambient. |
| $R_{\theta JC}$ | Thermal resistance junction to case. |
| $R_{\theta JSP(IS)}$ | Thermal resistance junction to solder pads (infinite sink mount to PCB). |

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, figure 2, and figure 3 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "UB" suffix can also be omitted. The radiation hardened designator M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I and table II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 2 | Optional | Optional |
| 3a<br>3b<br>3c | Required<br>Not applicable<br>Thermal impedance (transient), method 3131 of MIL-STD-750 (1) | Required<br>Not applicable<br>Thermal impedance (transient), method 3131 of MIL-STD-750 (1) |
| 4 | Required | Optional |
| 5 | Required | Not applicable |
| 8 | Required | Not required |
| 9 | $I_{CBO1}$, $h_{FE1}$ | Not applicable |
| 10 | 48 hours minimum | 48 hours minimum |
| 11 | $I_{CBO1}$; $h_{FE1}$;<br>$\Delta I_{CBO1}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE1}$ = ±15 percent | $I_{CBO1}$; $h_{FE1}$ |
| 12 | See 4.3.2 | See 4.3.2 |
| 13 | Subgroups 2 and 3 of table I herein; $\Delta I_{CBO1}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE1}$ = ±15 percent | Subgroup 2 of table I herein;<br>$\Delta I_{CBO1}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE1}$ = ±15 percent |
| 15 | Required | Not required |
| 16 | Required | Not required |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.



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4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum, $T_A$ ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). The thermal impedance limit used in screen 3c shall comply with the thermal impedance graph on figure 8, figure 9, figure 10, and figure 11 (less than or equal to the curve value at the same $t_H$ time) and/or shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of table I, subgroup 1, and 2 inspection only (table E-VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Delta requirements only apply to subgroups B4 and B5. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2 and 4.5.3 herein.

4.4.2.1 Group B inspection (JANS), table E-VIa of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = 10 V dc. |
| B5 | 1027 | $V_{CB}$ = 10 V dc; $P_D \geq$ 100 percent of maximum rated $P_T$ (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)<br><br>Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table E-VIa, adjust $T_A$ or $P_D$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hours minimum, sample size = 45, c = 0; adjust $T_A$ or $P_D$ to achieve a $T_J$ = +225°C minimum. |



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4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Delta requirements shall be in accordance with table I, subgroup 2 and 4.5.3 herein; delta requirements only apply to subgroup C6.

4.4.3.1 Group C inspection (JANS), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; (not applicable for UB devices). |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and applied thermal impedance curves. |
| C6 | 1026 | 1,000 hours, $V_{CB}$ = 10 V dc, power and ambient temperature shall be applied to the device to achieve $T_J$ = +150°C minimum, and minimum power dissipation of 75 percent of max rated $P_T$ (see 1.3 herein); n = 45, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |



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4.4.3.2 Group C inspection (JAN, JANTX, and JANTXV), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; not applicable for UB devices. |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and applied thermal impedance curves. |
| C6 | | Not applicable. |

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2 herein), except group D, subgroup 2 may be performed separate from other subgroups. Group D inspection may also be performed ahead of the screening lot using die selected in accordance with MIL-PRF-19500 and related documents. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

* 4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta measurements shall be in accordance with the applicable steps of 4.5.3.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Delta requirements. Delta requirements shall be as specified below:

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | Unit |
|---|---|---|---|---|---|---|
| | | Method | Conditions | | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 40 V dc | $\Delta I_{CB01}$ (1) | 100 percent of initial value or 8 nA dc, whichever is greater. | |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = 1 V dc; $I_C$ = 100 mA dc; pulsed see 4.5.1 | $\Delta h_{FE1}$ (1) | ±25 percent change from initial reading. | |

(1) Devices which exceed the table I limits for this test shall not be accepted.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical Examination 3/ | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ 4/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ 5/ | 1022 | n = 15 devices, c = 0 | | | | |
| Temp cycling 3/ 4/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Hermetic seal 4/ 6/ | 1071 | n = 22 devices, c = 0 | | | | |
| Fine leak | | | | | | |
| Gross leak | | | | | | |
| Electrical measurements 4/ | | Table I, subgroup 2 | | | | |
| Bond strength 3/ 4/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hours or $T_A$ = +300°C at t = 2 hours n = 11 wires, c = 0 | | | | |
| Decap internal visual (design verification) 4/ | 2075 | n = 4 devices, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 7/ | 3131 | See 4.3.3 | $Z_{\theta JX}$ | | | °C/W |
| Collector to base cutoff current | 3036 | Condition D. $V_{CB}$ = 40 V dc, $I_E$ = 0 mA dc | $I_{CBO1}$ | | 100 | nA dc |
| Emitter to base cutoff current | 3061 | Condition D. $V_{(BR)EBO}$ = 4 V dc | $I_{EBO}$ | | 10 | nA dc |
| Collector to base cutoff current | 3036 | Bias condition D; $I_E$ = 0 mA dc | $I_{CBO2}$ | | | µA dc |
| 2N3724, L, 2N3724UB | | $V_{CB}$ = 50 V dc | | | 10 | |
| 2N3725, L, 2N3725UB | | $V_{CB}$ = 80 V dc | | | 10 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc; $I_C$ = 100 mA dc | $h_{FE1}$ | | | |
| 2N3724, L, 2N3724UB | | | | 60 | 240 | |
| 2N3725, L, 2N3725UB | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 300 mA dc | $h_{FE2}$ | | | |
| 2N3724, L, 2N3724UB | | | | 40 | | |
| 2N3725, L, 2N3725UB | | | | 35 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 500 mA dc | $h_{FE3}$ | | | |
| 2N3724, L, 2N3724UB | | | | 35 | | |
| 2N3725, L, 2N3725UB | | | | 30 | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 1 A dc; pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N3724, L, 2N3724UB | | | | 30 | | |
| 2N3725, L, 2N3725UB | | | | 25 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $h_{FE5}$ | 30 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 800 mA dc; pulsed (see 4.5.1) | $h_{FE6}$ | | | |
| 2N3724, L, 2N3724UB | | | | 25 | | |
| 2N3725, L, 2N3725UB | | | | 20 | | |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 10 mA dc; $I_B$ = 1.0 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.25 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.25 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 100 mA dc; $I_B$ = 10 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.2 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.26 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 300 mA dc; $I_B$ = 30 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)3}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.32 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.4 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)4}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.42 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.52 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 800 mA dc; $I_B$ = 80 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)5}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.65 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.8 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = 1 A dc; $I_B$ = 100 mA dc; pulsed (see 4.5.1) | $V_{CE(sat)6}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.75 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.95 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 10 mA dc; $I_B$ = 1 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.76 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.76 | V dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 100 mA dc; $I_B$ = 10 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.86 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.86 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 300 mA dc; $I_B$ = 30 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)3}$ | | 1.1 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 500 mA dc; $I_B$ = 50 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)4}$ | .8 | 1.1 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 800 mA dc; $I_B$ = 80 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)5}$ | | 1.5 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A; $I_C$ = 1 A dc; $I_B$ = 100 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)6}$ | | 1.7 | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; | $I_{CBO3}$ | | | |
| 2N3724, L, 2N3724UB | | $V_{CB}$ = 80 V dc for 2N3725 | | | 120 | µA dc |
| 2N3725, L, 2N3725UB | | $V_{CB}$ = 50 V dc for 2N3724 | | | 120 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 800 mA dc | $h_{FE7}$ | | | |
| 2N3724, L, 2N3724UB | | | | 15 | | |
| 2N3725, L, 2N3725UB | | | | 10 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc, $I_C$ = 1 A dc | $h_{FE8}$ | | | |
| 2N3724, L, 2N3724UB | | | | 15 | | |
| 2N3725, L, 2N3725UB | | | | 10 | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Magnitude of small-signal short- circuit forward current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 50 mA dc; f = 100 MHz | $\lvert h_{fe} \rvert$ | 3.0 | | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 12 | pF |
| Input capacitance (output open- circuited) | 3240 | $V_{EB}$ = -0.5 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 55 | pF |
| Switching time | 3251 | | | | | |
| Turn-on delay time | | $V_{CC}$ = 30 V dc, $I_{CS}$ = 500 mA dc, $I_{B1}$ = 50 mA dc, $V_{BE\ (off)}$ = -3.8 V dc. See figure 12. | | | | |
| 2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | | | $t_{on}$ | | 50 | ns |
| Turn-off time | | $I_C$ = 500 mA dc, $I_{B1}$ = 50 mA dc, $I_{B2}$ = -50 mA dc. See figure 13. | $t_{off}$ | | 60 | ns |
| Subgroup 5 | | $T_A$ = +25°C | | | | |
| Safe operating area | 3051 | tp = 10 ms, (For TO-5 and TO-39 see figures 14 ad 15)<br>(For UB see figures 16 and 17) | | | | |
| Test 1 | | | | | | |
| 2N3724 2N3724L<br>2N3725, 2N3725L | | Vce = 7V, Ic = .8 A | | | | |
| 2N3724UB and 2N3725UB | | Vce = 11V, Ic = .5 A | | | | |
| Test 2 | | | | | | |
| 2N3724 2N3724L<br>2N3725, 2N3725L<br>2N3724UB and 2N3725UB | | Vce = 20 V, Ic = .25 A | | | | |
| Test 3 | | | | | | |
| 2N3724 2N3724L,<br>2N3724UB | | Vce = 30 V, Ic = .11 A | | | | |
| 2N3725, 2N3725L<br>2N3725UB | | Vce = 50 V, Ic = .04 A | | | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 6<br><br>Not required | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed in table I, subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ This hermetic seal test is an end-point to temp cycling in addition to electrical measurements.
* 7/ For end-point measurements, this test is required for the following subgroups:
Group B, step 1 of 4.4.2.2 herein (JAN, JANTX, and JANTXV).
Group B, subgroups 3, 4, and 5 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1 and 2.



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TABLE II. Group D inspection and end-point limits.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Neutron irradiation | 1017 | Neutron exposure $V_{CES}$ = 0 V | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 40 Vdc, $I_E$ = 0 mA dc | $I_{CBO1}$ | | 200 | ηA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{(BR)EBO}$ = 4 V dc | $I_{EBO}$ | | 20 | ηA dc |
| Collector to base cutoff current<br>2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | 3036 | Bias condition D; $I_E$ = 0 mA dc<br>$V_{CB}$ = 50 Vdc<br>$V_{CB}$ = 80 Vdc | $I_{CBO2}$ | | <br>20<br>20 | <br>µA dc<br>µA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 100 mA dc | $h_{FE1}$ | [30] | 240 | |
| Forward-current transfer ratio<br>2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 300 mA dc | $h_{FE2}$ | <br>[20]<br>[17.5] | | |
| Forward-current transfer ratio<br>2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 500 mA dc | $h_{FE3}$ | <br>[17.5]<br>[15] | | |
| Forward-current transfer ratio<br>2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | 3076 | $V_{CE}$ = 5 V dc, $I_C$ = 1.0 A dc; pulsed (see 4.5.1) | $h_{FE4}$ | <br>[15]<br>[12.5] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $h_{FE5}$ | [15] | | |
| Forward-current transfer ratio<br>2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | 3076 | $V_{CE}$ = 2 V dc, $I_C$ = 800 mA dc; pulsed (see 4.5.1) | $h_{FE6}$ | <br>[12.5]<br>[10] | | |
| Collector-emitter saturated voltage<br>2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | 3071 | $I_C$ = 10 mA dc, $I_B$ = 1.0 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | <br>0.29<br>0.29 | <br>V dc<br>V dc |
| Collector-emitter saturated voltage<br>2N3724, L, 2N3724UB<br>2N3725, L, 2N3725UB | 3071 | $I_C$ = 100 mA dc, $I_B$ = 10 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | <br>0.23<br>0.30 | <br>V dc<br>V dc |

See footnotes at end of table.



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TABLE II. Group D inspection and end-point limits - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 - Continued. | | | | | | |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 300 mA dc, $I_B$ = 30 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)3}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.37 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.46 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 500 mA dc, $I_B$ = 50 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)4}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.48 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.60 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 800 mA dc, $I_B$ = 80 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)5}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.75 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.92 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 1.0 A dc, $I_B$ = 100 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)6}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.86 | V dc |
| 2N3725, L, 2N3725UB | | | | | 1.09 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 10 mA dc, $I_B$ = 1 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.87 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.87 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 100 mA dc, $I_B$ = 10 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.99 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.99 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 300 mA dc, $I_B$ = 30 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)3}$ | | 1.27 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 500 mA dc, $I_B$ = 50 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)4}$ | 0.8 | 1.27 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 800 mA dc, $I_B$ = 80 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)5}$ | | 1.73 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 1.0 A dc, $I_B$ = 100 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)6}$ | | 1.96 | V dc |

See footnote at end of table.



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TABLE II. Group D inspection and end-point limits - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 | | | | | | |
| Steady-state total dose irradiation | 1019 | | | | | |
| 2N3724, L, 2N3724UB | | Gamma exposure $V_{ECS}$ = 24 V | | | | |
| 2N3725, L, 2N3725UB | | Gamma exposure $V_{ECS}$ = 40 V | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 40 Vdc, $I_E$ = 0 mA dc | $I_{CBO1}$ | | 200 | ηA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{(BR)EBO}$ = 4 V dc | $I_{EBO}$ | | 20 | ηA dc |
| Collector to base cutoff current | 3036 | Bias condition D; $I_E$ = 0 mA dc | $I_{CBO2}$ | | | |
| 2N3724, L, 2N3724UB | | $V_{CB}$ = 50 Vdc | | | 20 | μA dc |
| 2N3725, L, 2N3725UB | | $V_{CB}$ = 80 Vdc | | | 20 | μA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 100 mA dc | $h_{FE1}$ | [30] | 240 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 300 mA dc | $h_{FE2}$ | | | |
| 2N3724, L, 2N3724UB | | | | [20] | | |
| 2N3725, L, 2N3725UB | | | | [17.5] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 500 mA dc | $h_{FE3}$ | | | |
| 2N3724, L, 2N3724UB | | | | [17.5] | | |
| 2N3725, L, 2N3725UB | | | | [15] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc, $I_C$ = 1.0 A dc; pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N3724, L, 2N3724UB | | | | [15] | | |
| 2N3725, L, 2N3725UB | | | | [12.5] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1 V dc, $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $h_{FE5}$ | [15] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc, $I_C$ = 800 mA dc; pulsed (see 4.5.1) | $h_{FE6}$ | | | |
| 2N3724, L, 2N3724UB | | | | [12.5] | | |
| 2N3725, L, 2N3725UB | | | | [10] | | |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 10 mA dc, $I_B$ = 1.0 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.29 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.29 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 100 mA dc, $I_B$ = 10 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.23 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.30 | V dc |

See footnotes at end of table.



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TABLE II. Group D inspection and end-point limits - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued. | | | | | | |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 300 mA dc, $I_B$ = 30 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)3}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.37 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.46 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 500 mA dc, $I_B$ = 50 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)4}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.48 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.60 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 800 mA dc, $I_B$ = 80 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)5}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.75 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.92 | V dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 1.0 A dc, $I_B$ = 100 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)6}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.86 | V dc |
| 2N3725, L, 2N3725UB | | | | | 1.09 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 10 mA dc, $I_B$ = 1 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.87 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.87 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 100 mA dc, $I_B$ = 10 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | | |
| 2N3724, L, 2N3724UB | | | | | 0.99 | V dc |
| 2N3725, L, 2N3725UB | | | | | 0.99 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 300 mA dc, $I_B$ = 30 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)3}$ | | 1.27 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 500 mA dc, $I_B$ = 50 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)4}$ | 0.8 | 1.27 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 800 mA dc, $I_B$ = 80 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)5}$ | | 1.73 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 1.0 A dc, $I_B$ = 100 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)6}$ | | 1.96 | V dc |

1/ For sampling plan see MIL-PRF-19500.



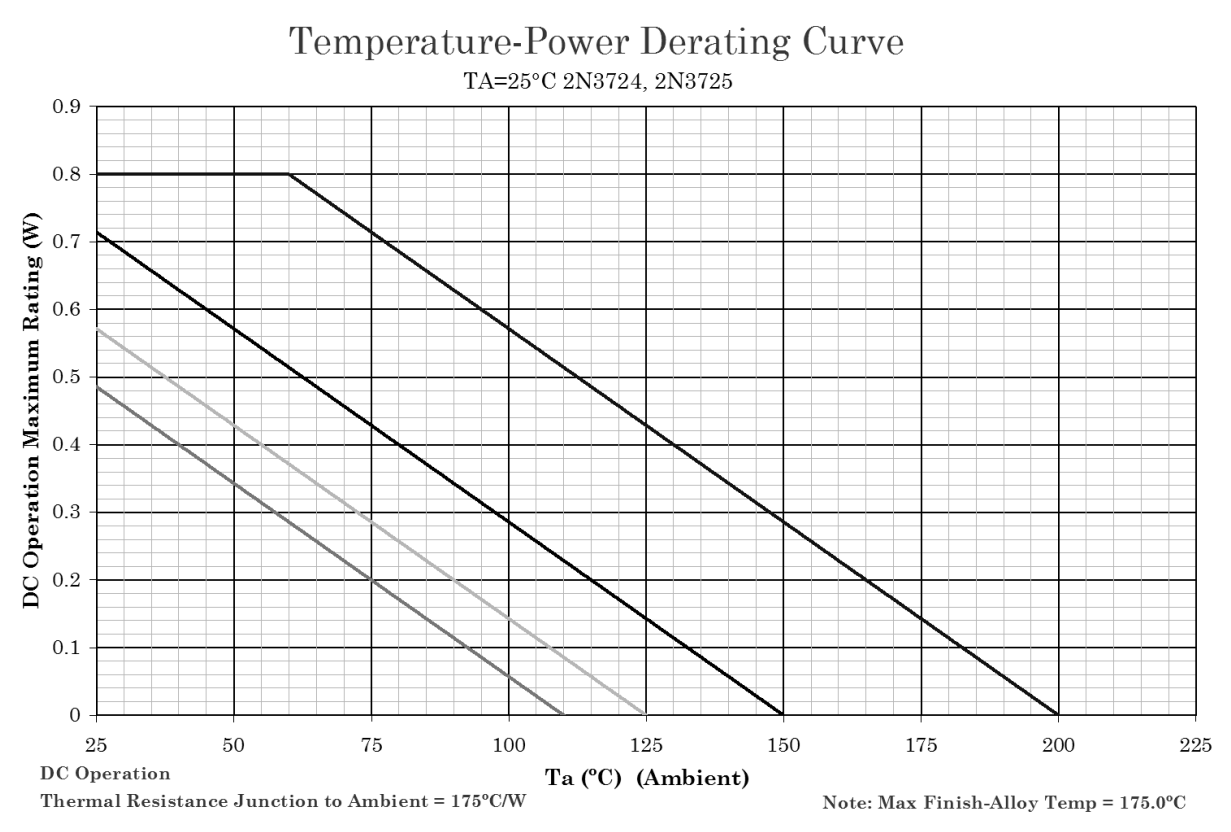
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* TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.3 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.3 herein. | |
| Subgroup 4 | | | |
| Thermal resistance | 3131 | $R_{\theta JSP(IS)}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets).<br>$R_{\theta JA}$, $R_{\theta JC}$ only. | 15 devices,<br>c = 0 |
| Thermal impedance curves | | See MIL-PRF-19500, table E-IX, group E, subgroup 4. | |
| Subgroup 5 | | | |
| Not applicable | | | |
| * Subgroup 6 | | | 11 devices |
| ESD | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B. | |



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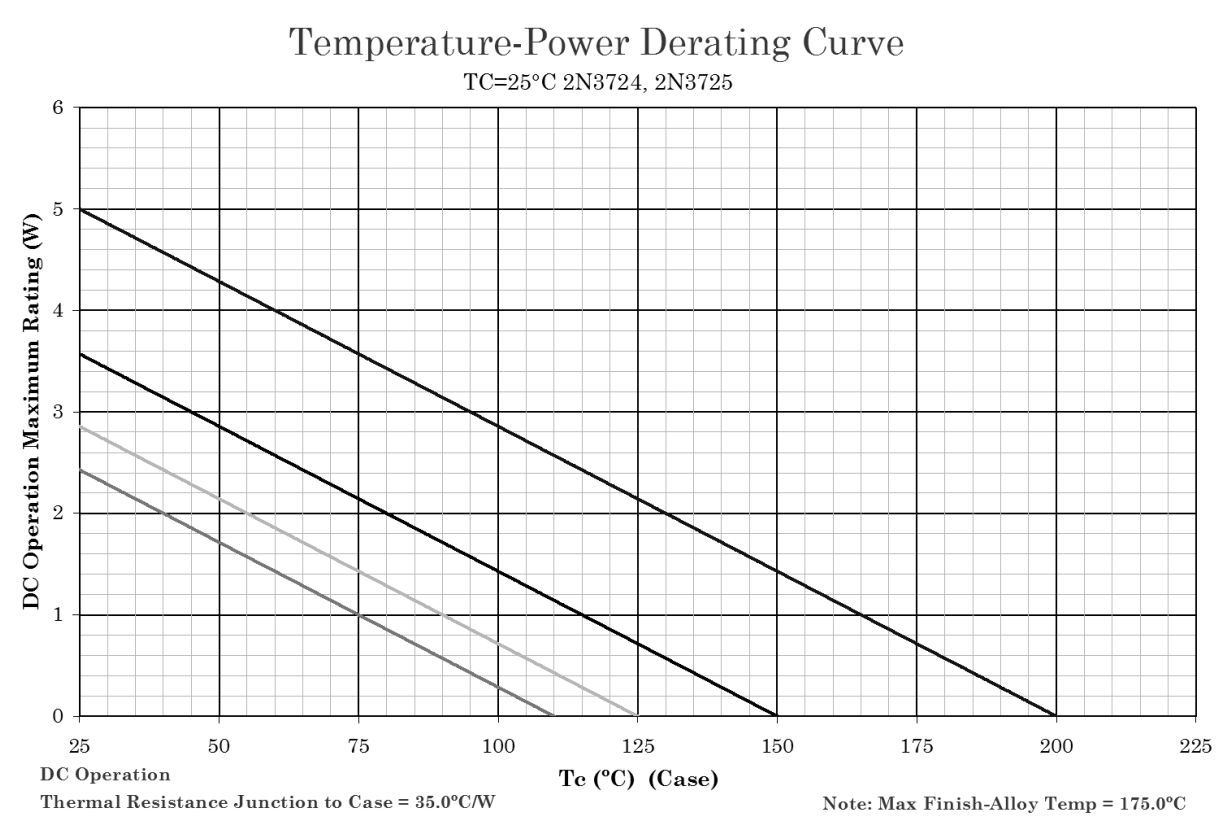

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 4. Derating for 2N3724 and 2N3725 ($R_{\theta JA}$) .125 inch (3.175 mm) PCB (TO-5 and TO-39).



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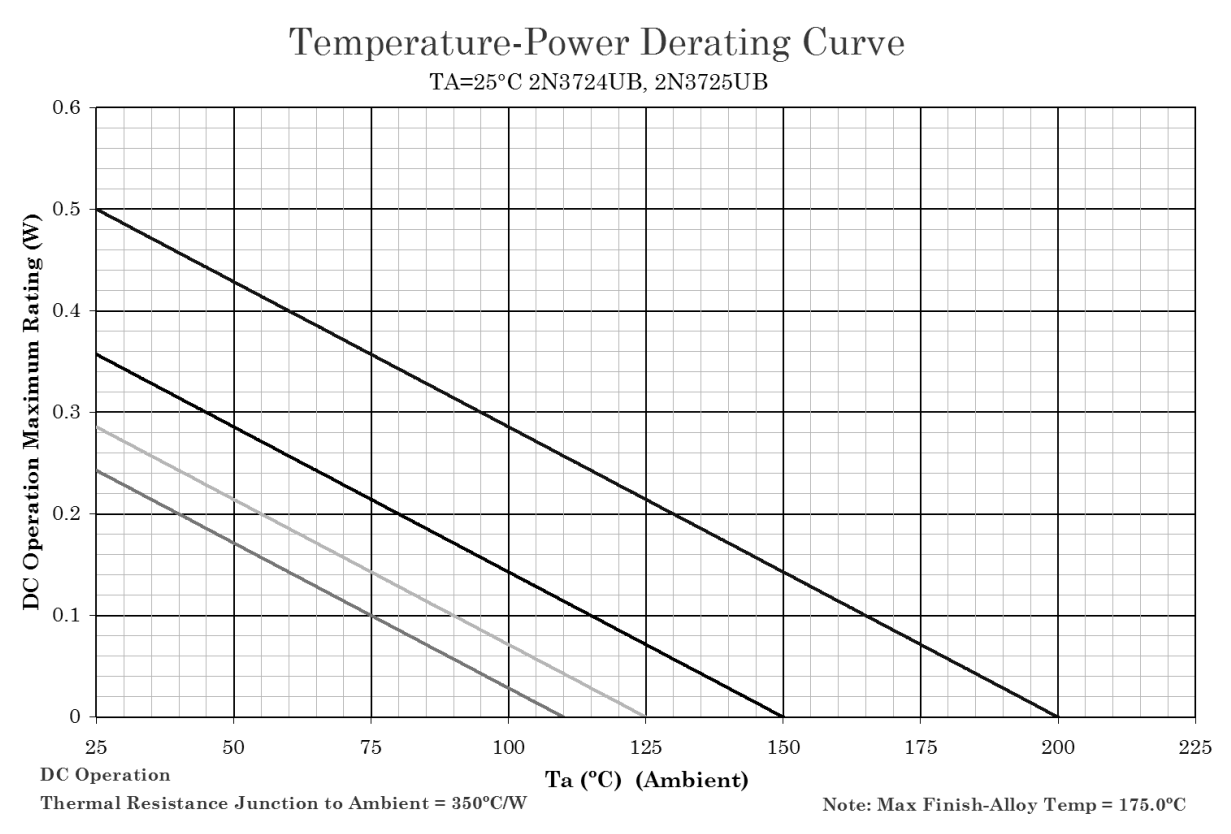

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤, 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 5. Derating for 2N3724 and 2N3725 ($R_{\theta JC}$), (TO-5 and TO-39).



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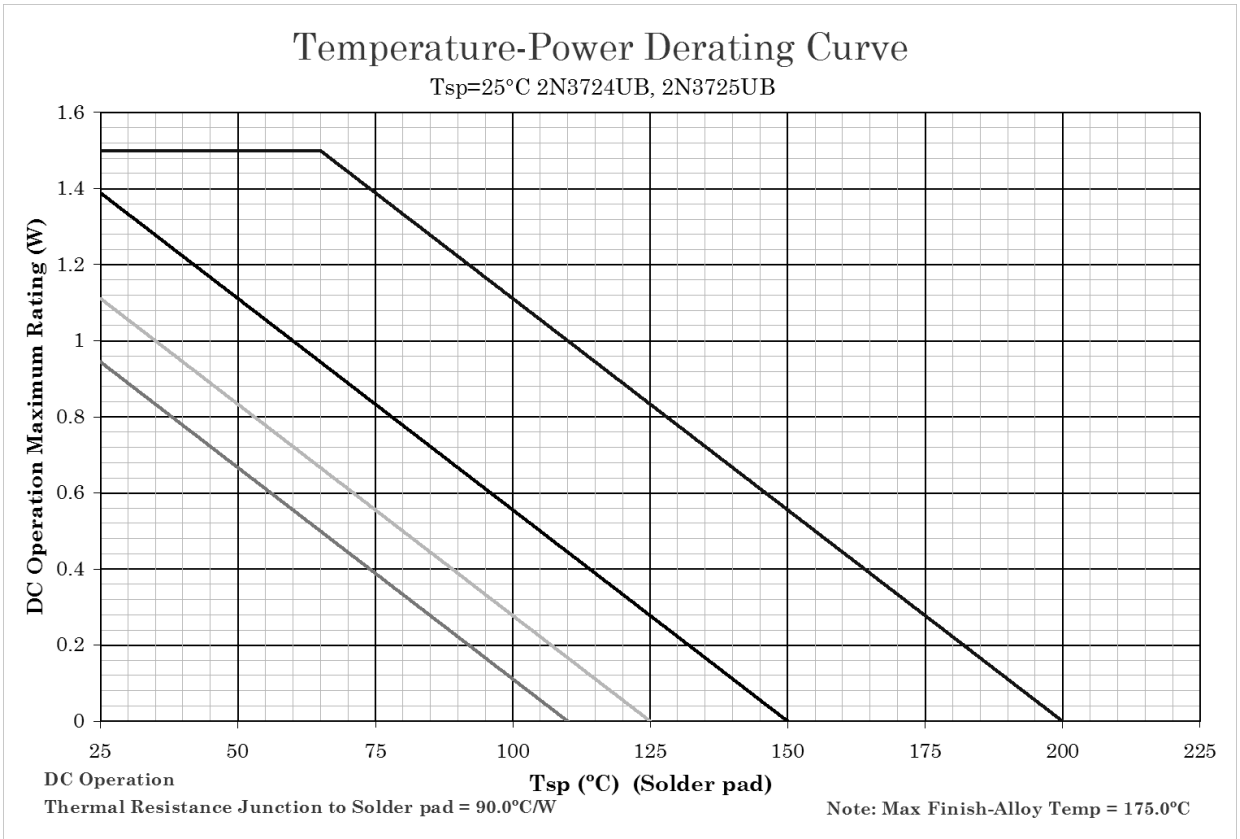

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 6. Derating for 2N3724UB and 2N3725UB ($R_{\theta JP(IS)}$).



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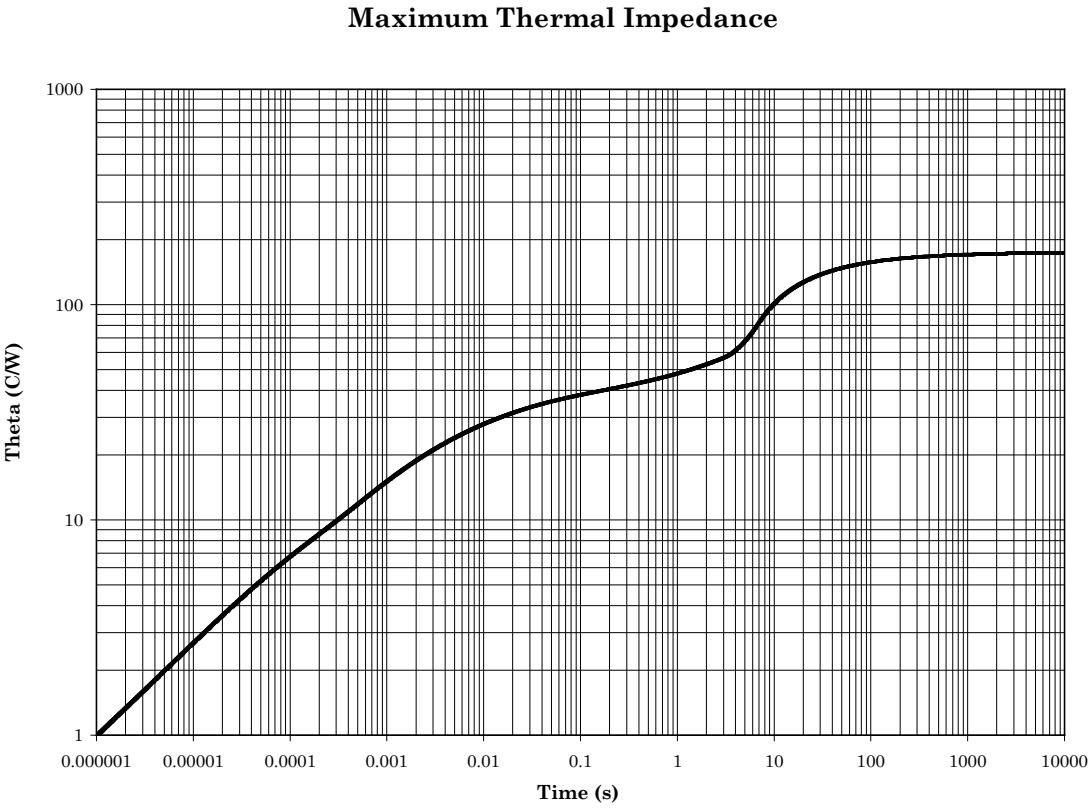

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤, 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 7. Derating for 2N3724UB and 2N3725UB ($R_{\theta JSP}$).



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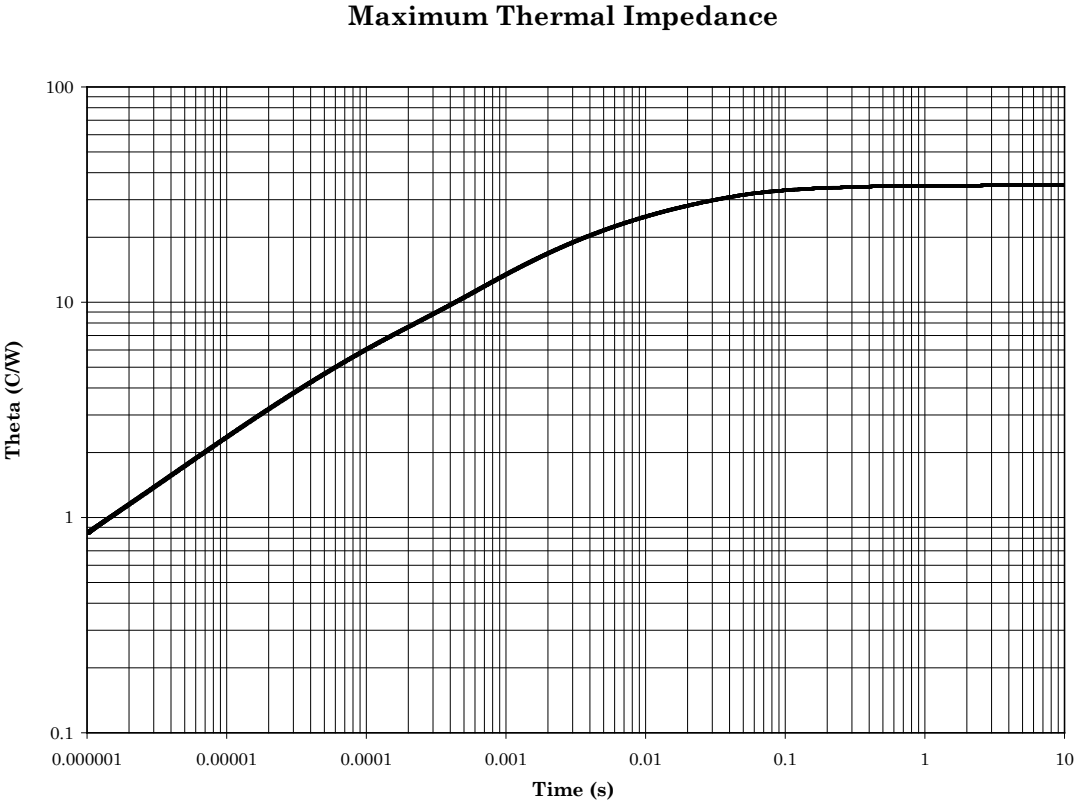



Ambient free air cooled $T_A$ = +25°C, 800mW, Thermal Resistance $R_{\theta JA}$ = 175°C/W.

FIGURE 8. Thermal impedance graph ($R_{\theta JA}$) for 2N3724 and 2N3725 (TO-5 and TO-39).



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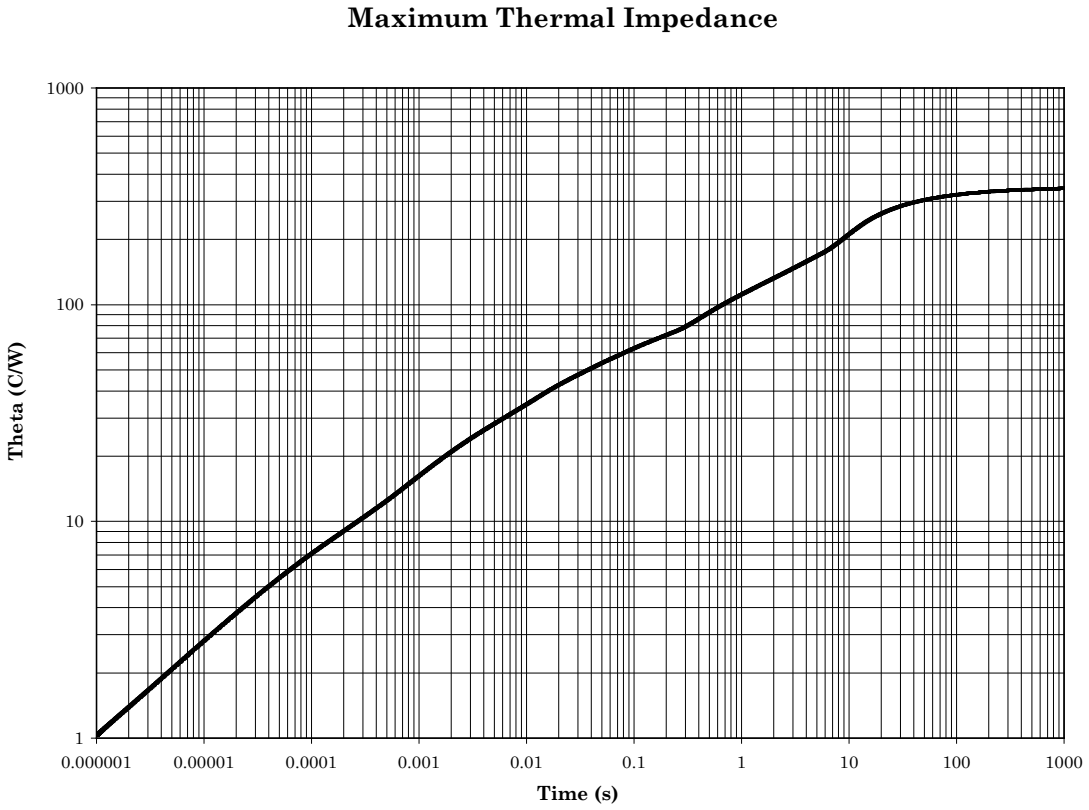



Ambient Case Mounted $T_C$ = +25°C, Thermal resistance $R_{\theta JC}$ = 35°C/W.

FIGURE 9. Thermal impedance graph ($R_{\theta JC}$) for 2N3724 and 2N3725 (TO-5 and TO-39).



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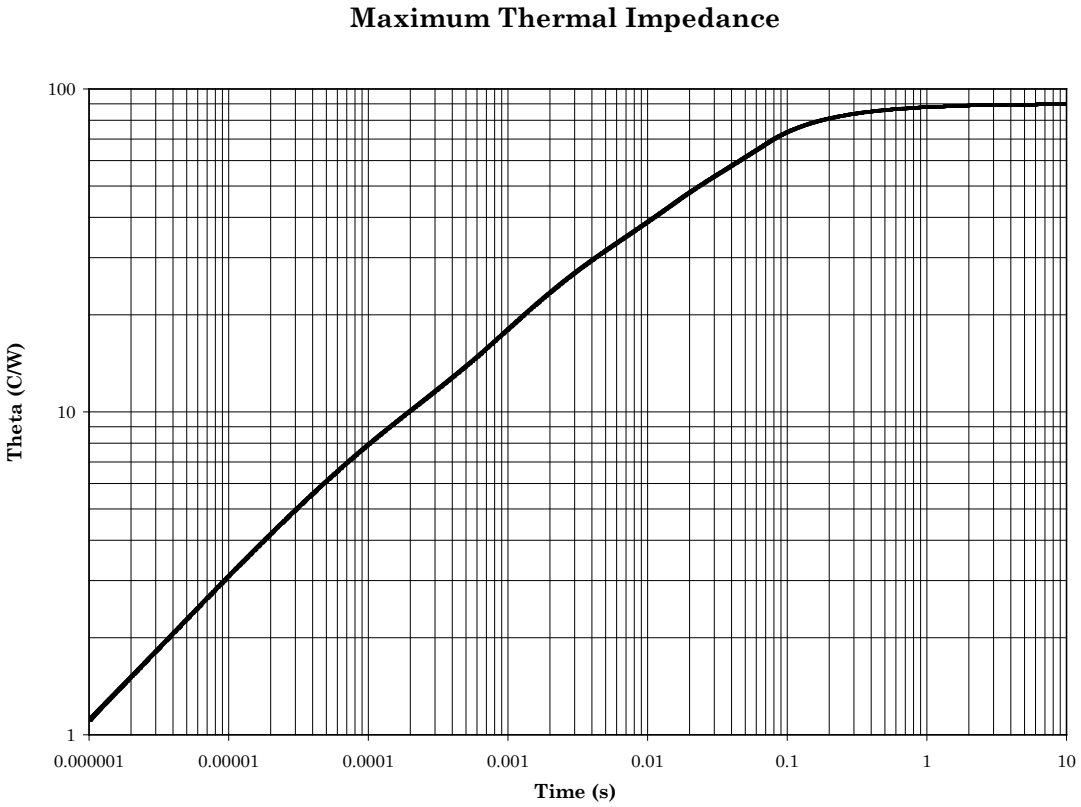



Mounted to minimal copper clad PCB at $T_A$ = +25°C (no heat sinking except air). Thermal resistance $R_{\theta JA}$ = 350°C/W

FIGURE 10. Thermal impedance graph ($R_{\theta JA}$) for 2N3724UB and 2N3725UB (UB).



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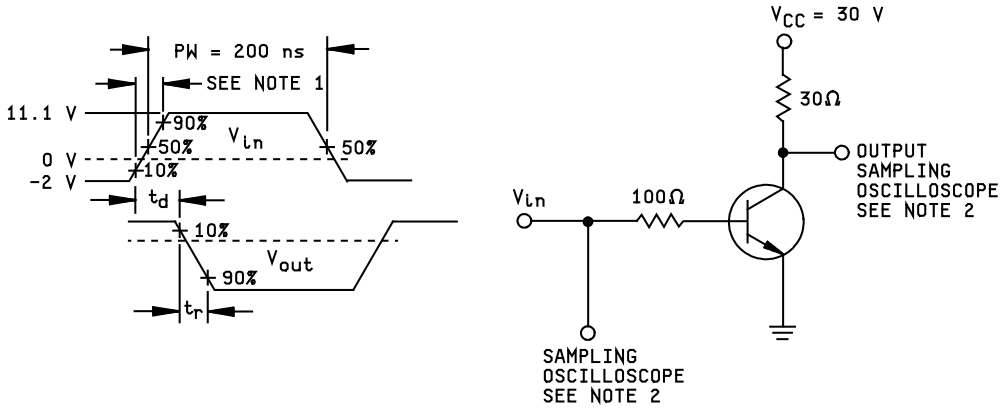



Solder mounted to very heavy copper clad PCB at $T_{\theta JSP(IS)}$ = +25°C. Thermal Resistance $R_{\theta JSP(IS)}$ = 90°C/W

FIGURE 11. Thermal impedance graph ($R_{\theta JSP(IS)}$) for 2N3724UB and 2N3725UB (UB).

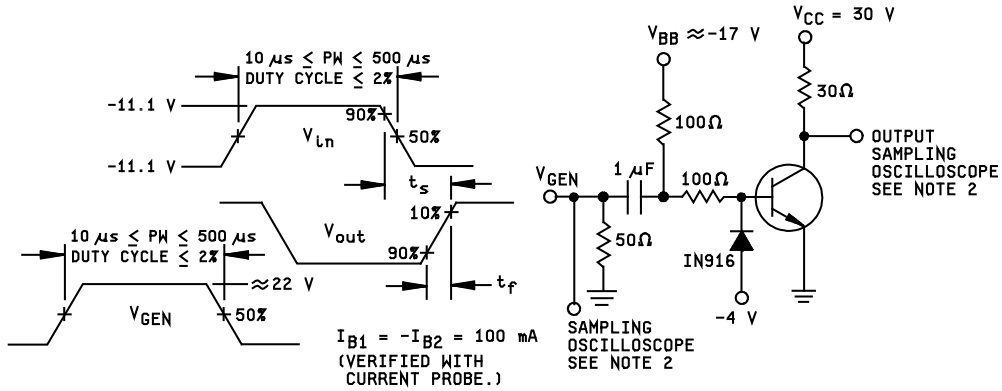


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NOTES:
1. The rise time ($t_r$) of the applied pulse shall be ≤ 0.1 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ, $C_{in}$ ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 12. Test circuit and waveforms for measuring turn-on.




NOTES:
1. The rise time ($t_r$) of the applied pulse shall be ≤ 0.1 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: $Z_{in}$ ≥ 100 kΩ, $C_{in}$ ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 13. Test circuit and waveforms for measuring turn-off.



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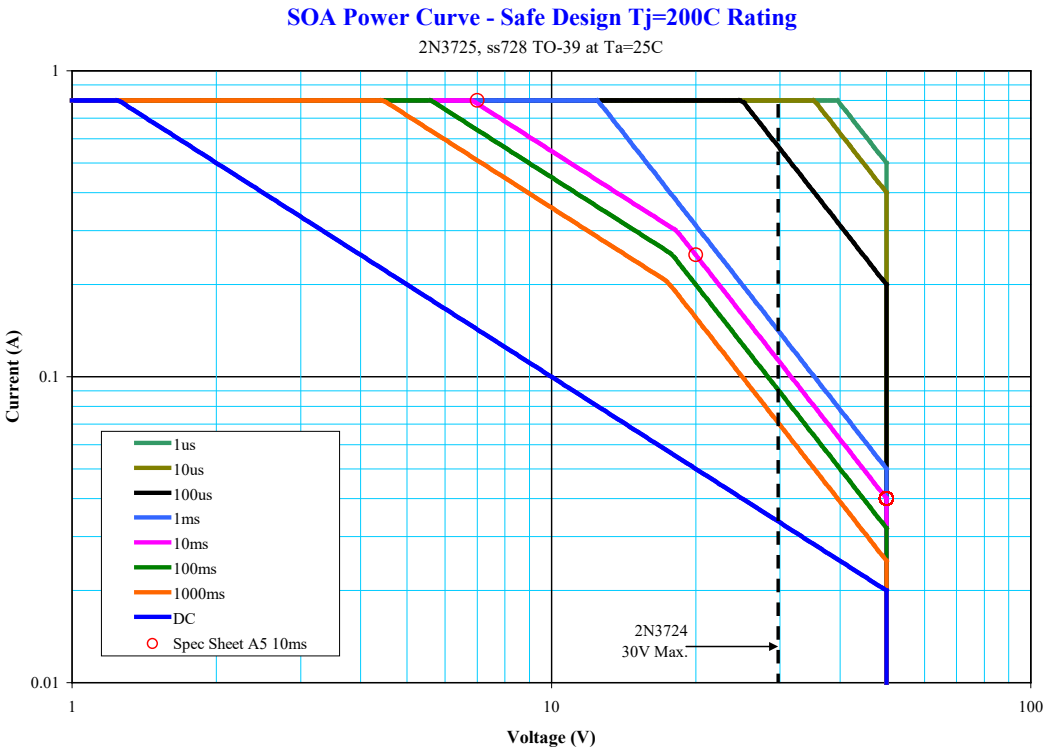



FIGURE 14. Maximum safe operating area.



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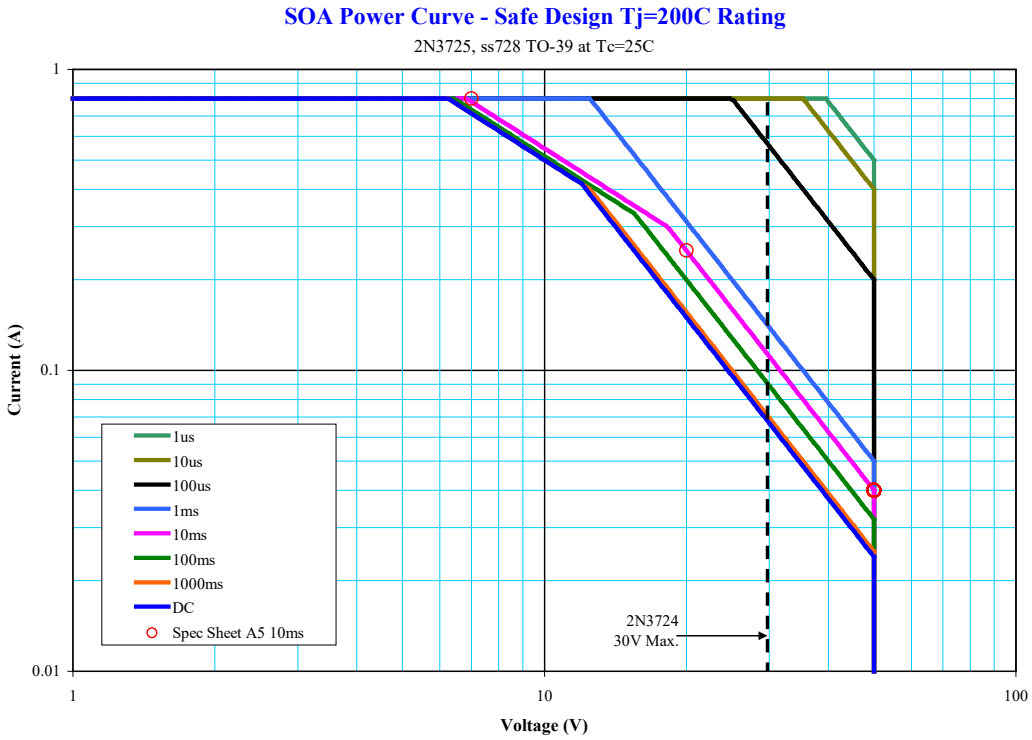



FIGURE 15. Maximum safe operating area.



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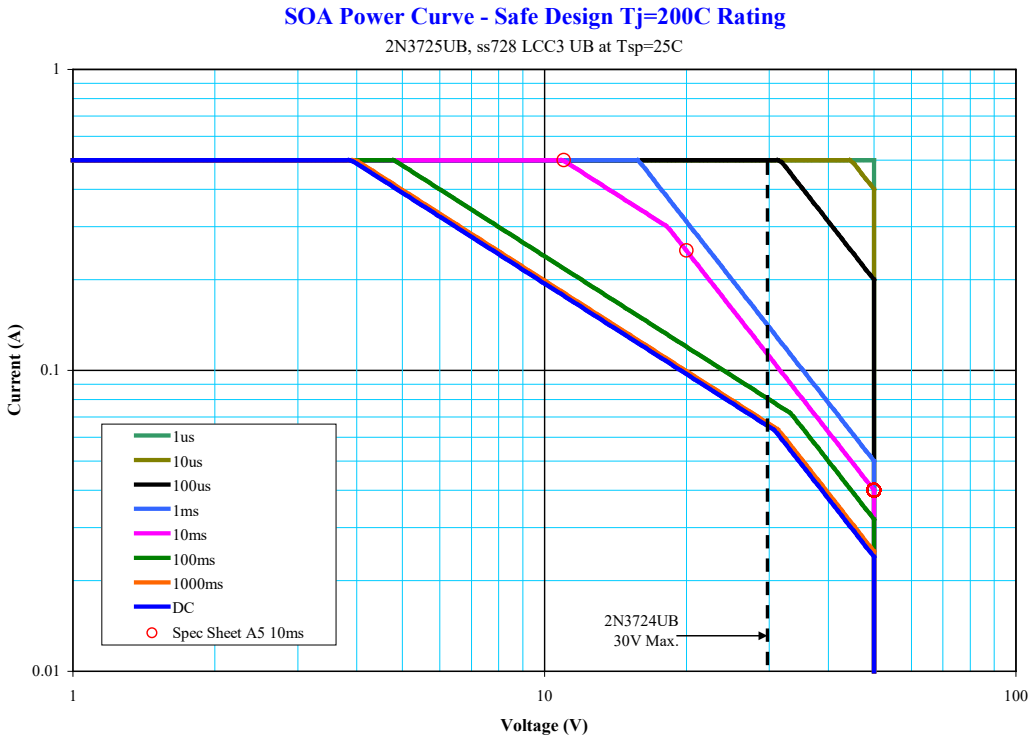



FIGURE 16. Maximum safe operating area.



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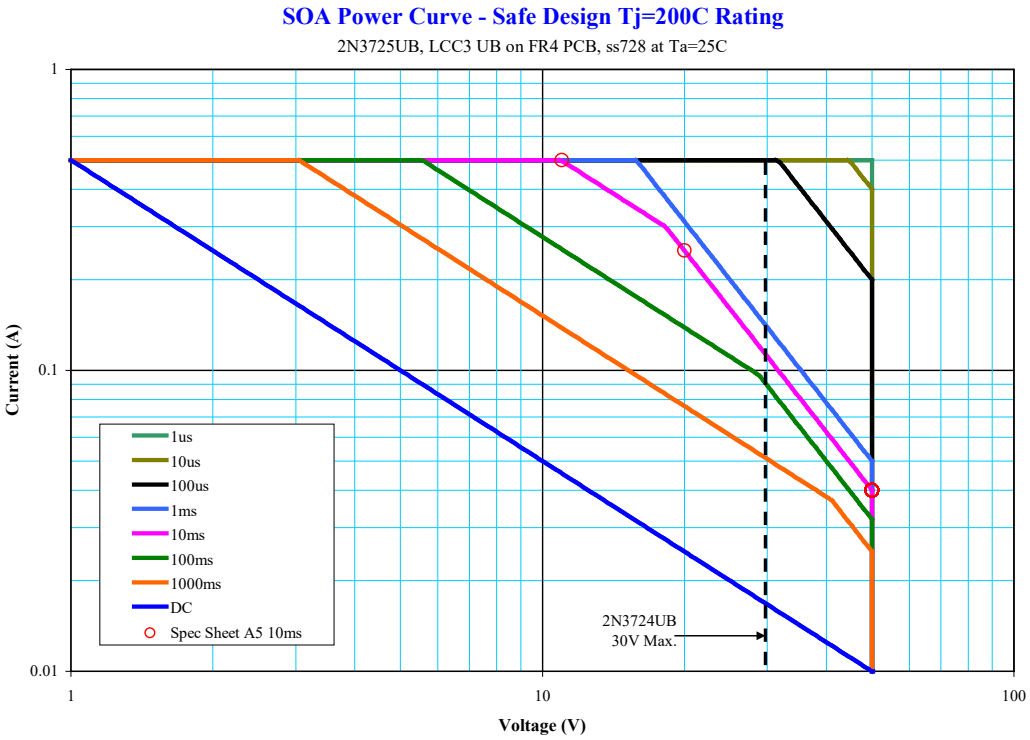



FIGURE 17. Maximum safe operating area.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

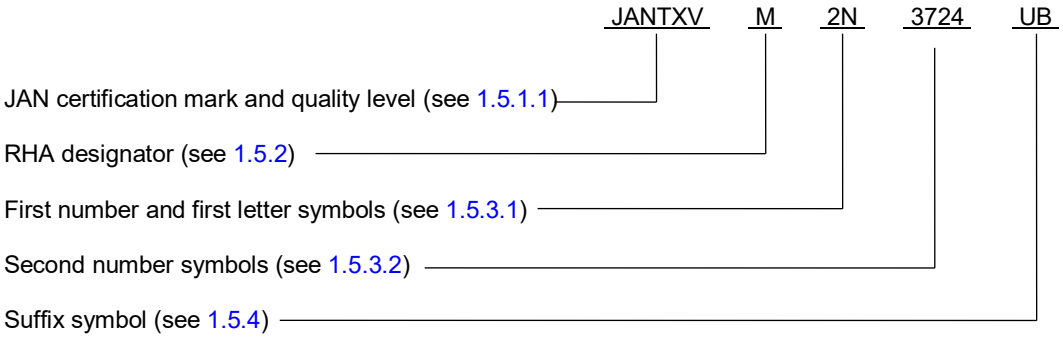
* d. The complete Part or Identifying Number (PIN), see 1.5 and 6.4.

e. For acquisition of RHA designed devices, Table II, subgroup 1 testing of Group D is optional. If subgroup 1 testing is desired, it should be specified in the contract.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

* 6.4 PIN construction example.

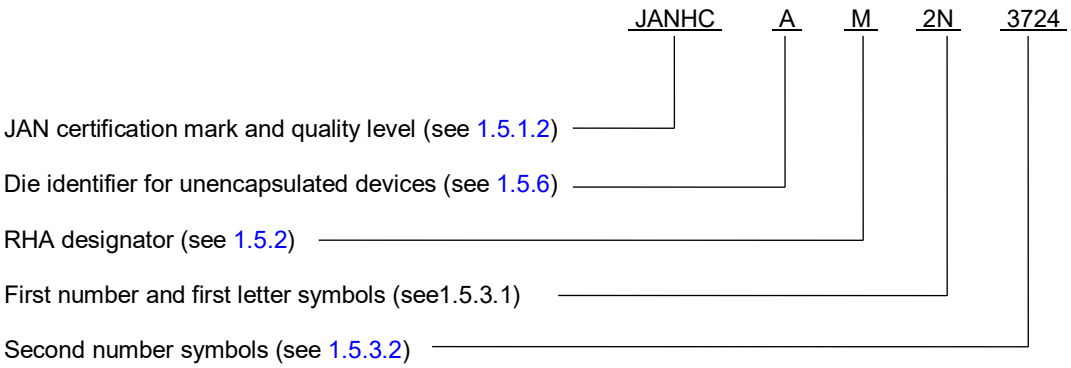
* 6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.






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* 6.4.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.




* 6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

* 6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for types 2N3724 and 2N3725. | | | |
|---|---|---|---|
| JAN1N3724<br>JAN1N3724L<br>JAN1N3724UB | JANTX1N3724<br>JANTX1N3724L<br>JANTX1N3724UB | JANTXV1N3724<br>JANTXV1N3724L<br>JANTXV1N3724UB | JANS1N3724<br>JANS1N3724L<br>JANS1N3724UB |
| JAN1N3725<br>JAN1N3725L<br>JAN1N3725UB | JANTX1N3725<br>JANTX1N3725L<br>JANTX1N3725UB | JANTXV1N3725<br>JANTXV1N3725L<br>JANTXV1N3725UB | JANS1N3725<br>JANS1N3725L<br>JANS1N3725UB |

* 6.5.2 List of PINs and suppliers for unencapsulated devices. The following is a list of possible PINs for unencapsulated devices available on this specification sheet.

| Die ordering information (1) | |
|---|---|
| PIN | Manufacturer |
| | 43611 |
| 2N3724 | JANHCA#2N3724<br>JANKCA#2N3724 |
| 2N3725 | JANHCA#2N3725<br>JANKCA#2N3725 |

* (1) The number sign (#) represent one of six RHA designators available (M, D, P, L, R, or F). The PIN is also available without a RHA designator.



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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
* Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2019-097)

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